8. If two dice are rolled, then find the probability of getting the product of face value 6 or the difference of face values 5.
9. In a two children family, find the probability that there is at least one girl in a family.
10. A bag contains 5 white and some black balls. If the probability of drawing a black ball from the bag is twice the probability of drawing a white ball then find the number of black balls.
11. The probability that a student will pass the final examination in both English and Tamil is 0.5 and the probability of passing neither is 0.1. If the probability of passing the English examination is 0.75, what is the probability of passing the Tamil examination?
12. The King, Queen and Jack of the suit spade are removed from a deck of 52 cards. One card is selected from the remaining cards. Find the probability of getting (i) a diamond (ii) a queen (iii) a spade (iv) a heart card bearing the number 5.

Points to Remember

- Range $= L–S$ (L - Largest value, S - Smallest value)
- Coefficient of range $= \frac{L-S}{L+S}$; Variance $\sigma^2 = \frac{\sum_{i=1}^{n}(x_i-\overline{x})^2}{n}$
- Standard deviation $\sigma = \sqrt{\frac{\Sigma\left(x_i-\overline{x}\right)^2}{n}}$
- Standard deviation (ungrouped data)

  (i) Direct method $\sigma = \sqrt{\frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2}$ (ii) Mean method $\sigma = \sqrt{\frac{\Sigma d_i^{\,2}}{n}}$

  (iii) Assumed mean method $\sigma = \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2}$

  (iv) Step deviation method $\sigma = c \times \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2}$
- Standard deviation of first n natural numbers $\sigma = \sqrt{\frac{n^2-1}{12}}$
- Standard deviation (grouped data)

  (i) Mean method $\sigma = \sqrt{\frac{\Sigma f_i d_i^2}{N}}$ (ii)Assumed mean method $\sigma = \sqrt{\frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2}$

  (iii) Step deviation method $\sigma = C \times \sqrt{\frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2}$
- Coefficient of variation C.V $= \frac{\sigma}{\overline{x}} \times 100\%$
- If the C.V. value is less, then the observations of corresponding data are consistent. If the C.V. value is more then the observations of corresponding are inconsistent.

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